

Continuous Random Variable (CRV)

A continuous random variable is a type of random variable that can take any real value within a certain range or interval. Unlike discrete random variables, which take countable values (like 1, 2, 3...), continuous variables have uncountably infinite possibilities.

↳ So, For a random variable X with its range S_X consists of one or more interval:

if, X is continuous random variable:

then, $S_X \rightarrow$ contains infinite values

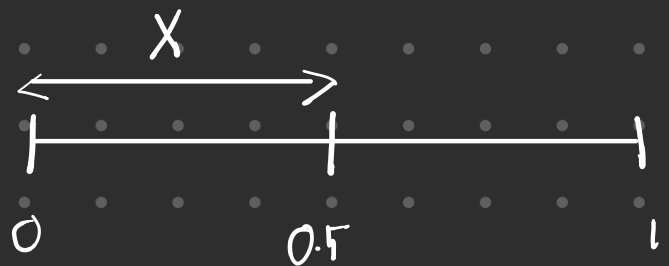
$$\therefore P[X=x] = \frac{1}{n} = \frac{1}{\infty} = 0$$

So, the probability of each individual is zero. (interesting...)

CDF of CRV

$$\hookrightarrow F_X(x) = P[X \leq x]$$

$$P[X \leq 0.5] = 0.5$$



$$P[0 \leq x \leq 0.5] = 0.5$$

$$P[0.25 \leq x \leq 0.5] = 0.25$$

* For any random variable X ,

i) $F_X(x_1) \leq F_X(x_2)$ for $x_1 < x_2$

ii) $F_X(+\infty) = 1$, $F_X(-\infty) = 0$

iii) $P[x_1 < X < x_2] = F_X(x_2) - F_X(x_1)$

* $P[a \leq X \leq b] = P[a < X \leq b] = P[a \leq X < b] = P[a < X < b]$
How?

$$P[a \leq X \leq b] \geq P[a < X \leq b]$$

$$\Rightarrow P[a \leq X \leq b] = P[a < X \leq b] + P[X = a]$$

$$\Rightarrow P[a \leq X \leq b] = P[a < X \leq b] + \underbrace{0}_{0} \rightarrow \frac{1}{\infty} = 0$$

$\hookrightarrow$ works other combinations too.

Example 2.10:

$$F_X(x) = \begin{cases} 0, & \text{for } x < 0 \\ \frac{x}{4}, & \text{for } 0 \leq x < 4 \\ 1, & \text{for } x \geq 4 \end{cases}$$

i) Draw the CDF curve.

ii) Find $P[2 < X \leq 3]$, $F_X(1.5)$

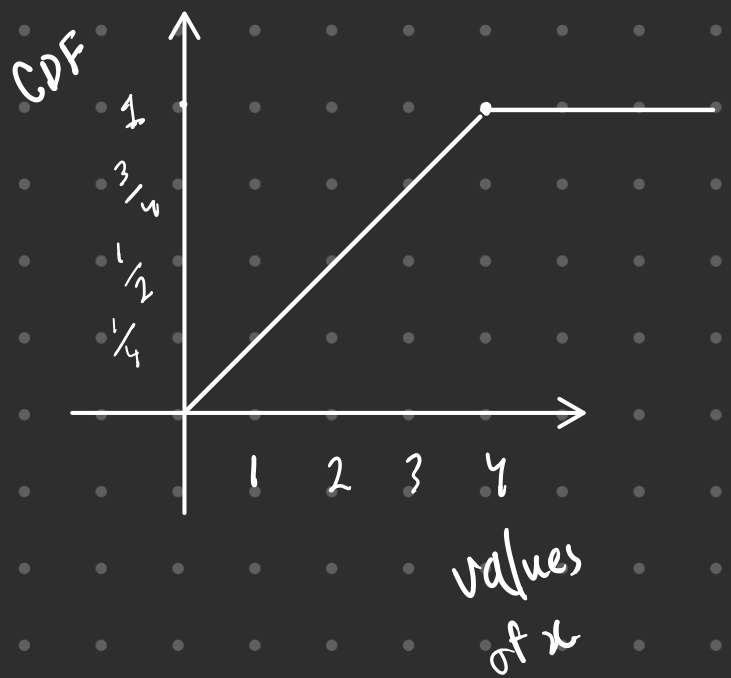
Solⁿ:

$$P[2 < x \leq 3]$$

$$= F_x(3) - F_x(2)$$

$$= \frac{3}{4} - \frac{1}{2} = \frac{1}{4}$$

$$F_x(1.5) = \frac{1.5}{4} = \frac{3}{8}$$



Probability Density Function (PDF)

Probability Density $\rightarrow$ Measure of the amount of probability per unit length.

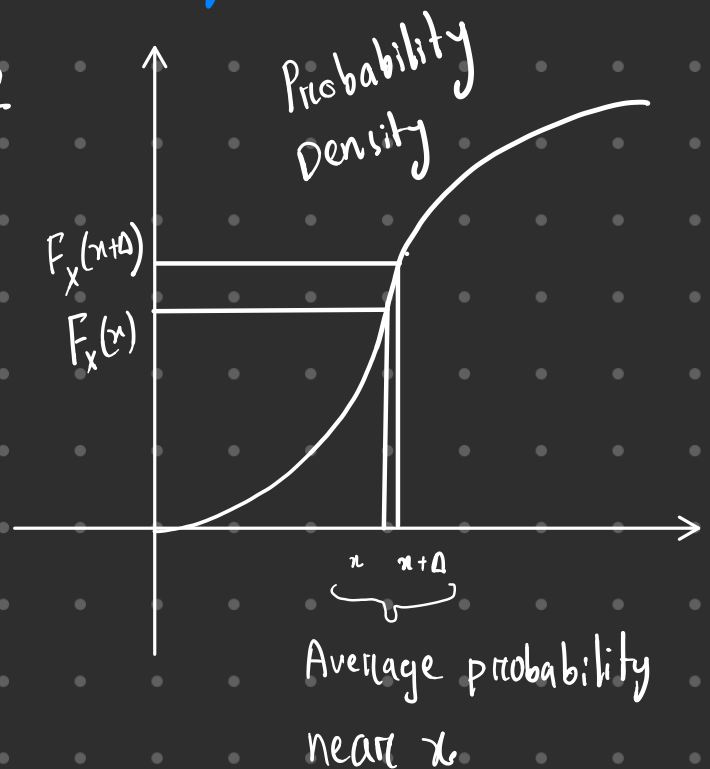
$$\hookrightarrow f_x(x) = \lim_{\Delta \rightarrow 0} \frac{F_x(x+\Delta) - F_x(x)}{\Delta}$$

PDF $\leftarrow$

$$= \left(\frac{P}{\Delta} \right)$$

$$\hookrightarrow f_x(x) = \frac{d}{dx} F_x(x)$$

$$\hookrightarrow F_x(x) = \int f_x(x)$$



$$P = F_x(x+\Delta) - F_x(x)$$
$$= \frac{F_x(x+\Delta) - F_x(x)}{\Delta} \cdot \Delta$$

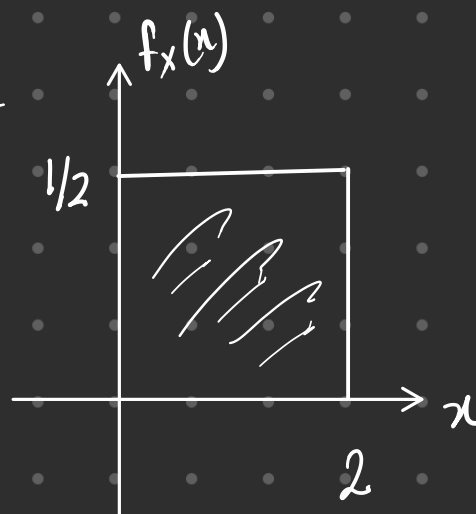
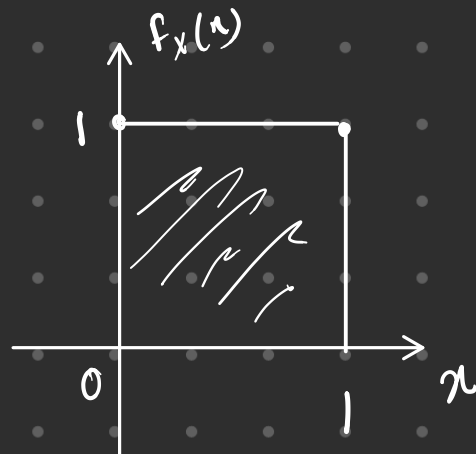
⊗ The PDF, denoted by $f_X(x)$, satisfies:

i) Non-negative: $f_X(x) \geq 0$

ii) Total Area = 1: $\int_{-\infty}^{+\infty} f_X(x) dx = 1$

iii) Interval Probability:

$$P(a \leq X \leq b) = \int_a^b f_X(x) dx$$



Example 2.11: $F_X(x) = \begin{cases} 0, & \text{for } x < 0 \\ \frac{x}{4}, & \text{for } 0 \leq x < 4 \\ 1, & \text{for } x \geq 4 \end{cases}$

i) Find the PDF from this and draw the curve.

ii) Find $P[2 < X \leq 3]$ using PDF.

Soln:

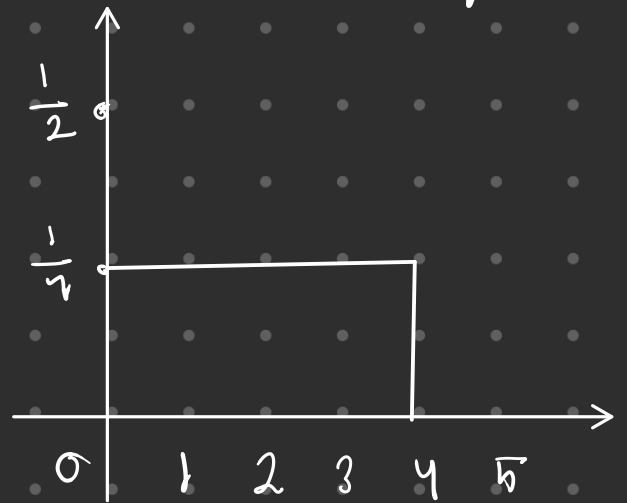
$$\frac{d}{dx}(0) = 0$$

$$\frac{d}{dx} \frac{x}{4} = \frac{1}{4}$$

$$\frac{d}{dx} 1 = 0$$

$$\leadsto f_X(x) = \begin{cases} \frac{1}{4}, & 0 \leq x < 4 \\ 0, & \text{otherwise} \end{cases}$$

$$P[2 < X \leq 3] = \int_2^3 \frac{1}{4} dx = \frac{3}{4} - \frac{1}{2} = \frac{1}{4}$$



Example 2.13: The PDF of a continuous random variable X is

$$f_X(x) = \begin{cases} cxe^{-x/2}, & \text{for } x \geq 0; \\ 0, & \text{otherwise.} \end{cases}$$

1. Find the value of the constant c .
2. Find the CDF of the random variable X .
3. Find the probability $P[2 \leq X \leq 5]$ from the PDF of X .
4. Find the probability $P[2 \leq X \leq 5]$ from the CDF of X .

Solⁿ:

$$\begin{aligned} \textcircled{1} \int_{-\infty}^{+\infty} f(x) dx &= \int_0^{+\infty} cxe^{-x/2} dx \\ &= cx \int_0^{+\infty} e^{-x/2} dx - c \int_0^{+\infty} x \left(\int_0^{+\infty} e^{-x/2} dx \right) dx \end{aligned}$$

$$\begin{aligned}
&= \left[Cx(-2)e^{-x/2} \right]_0^{+\infty} + 2C \int_0^{\infty} e^{-x/2} dx \\
&= (0-0) + 2C \left[-2e^{-x/2} \right]_0^{\infty} \\
&= 2C [0 - (-2)] = 4C
\end{aligned}$$

Here, $4C = 1 \Rightarrow C = \frac{1}{4}$.

② Given, $f_x(x) = \begin{cases} \frac{1}{4} x e^{-x/2} & \text{for } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$

CDF: $F_x(x) = \int_0^x f_x(x) dx$

$$= \int_0^x \frac{1}{4} x e^{-x/2} dx$$

$$= \frac{1}{4} \left[x \int_0^x e^{-x/2} dx - \int_0^x \left(\int_0^x e^{-x/2} dx \right) dx \right]$$

$$\begin{aligned}
&= \frac{1}{4} \left[x \cdot (-2)e^{-x/2} \right]_0^x - \\
&\quad \frac{1}{4} \int_0^x (-2)(e^{-x/2} - 1) dx
\end{aligned}$$

$$\begin{aligned}
&= \frac{1}{2} x (e^{-x/2} - 1) - \frac{1}{4} \int_0^x -2e^{-x/2} - \frac{1}{4} \int_0^x +2 dx \\
&= -\frac{1}{2} x (e^{-x/2} - 1) + \left[2(-2)e^{-x/2} \right]_0^x - \frac{1}{2} x \\
&\Rightarrow -\frac{1}{2} x e^{-x/2} + \frac{1}{2} x - e^{-x/2} + 1 - \frac{1}{2} x
\end{aligned}$$

$$\Rightarrow 1 - e^{-x/2} - \frac{1}{2} x e^{-x/2}$$

$$\therefore F_X(x) = \begin{cases} 1 - e^{-x/2} - \frac{1}{2} x e^{-x/2}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

$$\begin{aligned}
\textcircled{3} P[2 \leq X \leq 5] &= \int_2^5 \frac{1}{4} x e^{-x/2} dx \\
&= \left[1 - e^{-x/2} - \frac{1}{2} x e^{-x/2} \right]_2^5 = e^{-4} - e^{-10} = 0.448
\end{aligned}$$

$$\begin{aligned}
\textcircled{4} P[2 \leq X \leq 5] &= F_X(5) - F_X(2) \\
&= e^{-4} - e^{-10} = 0.448
\end{aligned}$$

(*) Expected value are well discussed in Discrete Random Variable

As for, Continuous Random Variable,

$$E[X] = \int_{-\infty}^{+\infty} x \cdot f_x(x) dx$$

$$V[X] = \int_{-\infty}^{+\infty} (x-\mu)^2 \cdot f_x(x) dx$$

Example:

Example: The time elapsed, in minutes, between the placement of an order of pizza and its delivery is random with pdf

$$f(x) = \begin{cases} 1/15 & \text{if } 25 < x < 40 \\ 0 & \text{otherwise.} \end{cases}$$

Solⁿ:

$$\text{Here, } E[X] = \int_{25}^{40} x \frac{1}{15} dx = 32.5$$

$$E[X^2] = \int_{25}^{40} x^2 \frac{1}{15} dx = 1075$$

$$\therefore V[X] = 1075 - (32.5)^2 = 18.75$$

$$\sigma_x = \sqrt{V[X]} = 4.33$$

Example

from slide:

Sketch the graph of the function

$$f(x) = \begin{cases} \frac{1}{2} - \frac{1}{4}|x-3| & 1 \leq x \leq 5 \\ 0 & \text{otherwise,} \end{cases}$$

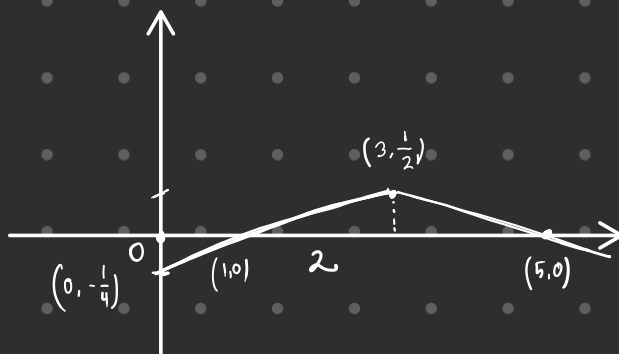
and show that it is the probability density function of a random variable X .

Find F , the distribution function of X , and show that it is continuous.

Find $E[X]$ and $V[X]$.

Soln:

Graph:



To check the validity of a function

to be PDF, it must be $\int_{-\infty}^{+\infty} f_X(x) dx = 1$

Here, $\int_1^5 f(x) dx$

$$= \int_1^5 \left(\frac{1}{2} - \frac{1}{4}|x-3| \right) dx = 1$$

So, it is a valid PDF.

Here, $f(x) = \frac{1}{2} - \frac{1}{4}|x-3|$, a piecewise linear.

$$\text{So, } f(x) = \begin{cases} 0, & \text{if } x < 1 \\ \frac{1}{2} + \frac{1}{4}(x-3), & 1 \leq x < 3 \\ \frac{1}{2} - \frac{1}{4}(x-3), & 3 \leq x < 5 \\ 0, & x \geq 5 \end{cases}$$

$$f(x) = 0 \Rightarrow$$

$$0 = \frac{1}{2} - \frac{1}{4}|x-3|$$

$$\frac{1}{4}|x-3| = \frac{1}{2}$$

$$\Rightarrow |x-3| = 2$$

$$\Rightarrow x-3 = 2 \quad -x+3 = 2$$

$$\Rightarrow x = 5 \quad \Rightarrow x = 1$$

$$x = 0 \Rightarrow$$

$$f(x) = \frac{1}{2} - \frac{1}{4}|x-3|$$

$$= \frac{1}{2} - \frac{3}{4} = -\frac{1}{4}$$

$$x = 3 \Rightarrow$$

$$f(x) = \frac{1}{2}$$

$$\text{For, } x < 1 \rightarrow F(x) = \int_{-\infty}^x 0 dx = 0$$

$$\text{For, } 1 \leq x < 3 \rightarrow \int_{-\infty}^x f(x) dx = \int_1^x \left[\frac{1}{2} + \frac{1}{4}(x-3) \right] dx$$

$$\rightarrow F(x) = \frac{1}{8}x^2 - \frac{1}{4}x + \frac{1}{8}$$

$$\begin{aligned} \text{For, } 3 \leq x < 5 \rightarrow \int_{-\infty}^x f(x) dx &= \int_1^3 f(x) dx + \int_3^x f(x) dx \\ &= F(3) + \int_3^x \left[\frac{1}{2} - \frac{1}{4}(x-3) \right] dx \\ &= \frac{1}{2} + \left(-\frac{1}{8}x^2 + \frac{5}{4}x - \frac{11}{8} \right) \\ &= -\frac{1}{8}x^2 + \frac{5}{4}x - \frac{17}{8} \end{aligned}$$

$$\text{For, } x \geq 5 \rightarrow F(x) = 1 \quad (\text{Beyond the interval})$$

$$\therefore F(x) = \begin{cases} 0 & x < 1 \\ \frac{1}{8}x^2 - \frac{1}{4}x + \frac{1}{8} & 1 \leq x \leq 3 \\ -\frac{1}{8}x^2 + \frac{5}{4}x - \frac{17}{8} & 3 < x \leq 5 \\ 1 & x > 5 \end{cases}$$

$$\text{Now, } E[X] = \int_1^5 x f(x) dx = 3$$

$$V[X] = E[X^2] - (E[X])^2 = \int_1^5 x^2 f(x) dx - 3^2 = \frac{2}{3}$$